The Norns III: Seeding Simulated-Annealing Placement with Provenance and Recovered Clusters How hot can the anneal start before it destroys a good initial placement?

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Abstract—Simulated annealing is the workhorse of FPGA placement, and it starts from an initial placement that the anneal then destroys and rebuilds. What if the initial placement carried real information – the HDL hierarchy a provenance-tracked front end kept, or the clusters a graph analysis recovered from the flat netlist? This paper asks whether such seeds improve placement, and, because the anneal's early phase can erase any seed, at what initial temperature the seed's advantage dies. The seeds come from Odin III's provenance hierarchy and from Loki III's L2 clusterings (Louvain, spectral, affinity propagation, k-means); the die-level arrangement of clusters comes from a cut-weighted quotient-graph layout; and the seeds enter VPR through its initial-placement and floorplanning-constraint interfaces, with VPR's stock checkpointing flag (on by default, following RLPlace) enabled throughout and an off sweep as the control arm. We report seed retention at scale 1 and at twice each design's crossover, quality of results against an unseeded baseline, and the largest tested temperature scale at which each seed still helps – the recipe a clustering-based initial placement should be used with.

Index Terms—FPGA placement, simulated annealing, provenance, clustering, VTR

I. Introduction

VPR's simulated-annealing placer starts from an initial placement and anneals it: the schedule begins hot enough to move anything, and the early phase rearranges the circuit wholesale. The question this paper asks is whether the initial placement can be made to mean something, and how much of that meaning survives the anneal. Gharibian et al. showed that system-level structure recovered from a flat netlist correlates with placement locality [1]–[3]. Odin III keeps the provenance link that makes recovery unnecessary for the hierarchy [1], and Loki III recovers clusters from the flat netlist without a language model. RLPlace [4] showed that the anneal itself can be improved – directed moves chosen by a reinforcement-learning agent and checkpointing that prevents oscillation – starting from VTR's analytic placement. This paper connects those lines: the seed carries the structure, the anneal is measured for how much of it the seed can keep, and the largest tested temperature scale at which the advantage survives is reported as data, not tuned away.

The contributions are: (1) a seeding methodology – provenance hierarchy and four recovered clusterings, arranged on the die by a quotient-graph layout, entering VPR through its initial-placement and constraint interfaces, with VPR's stock checkpointing enabled and its off sweep as the control arm; (2) the crossover sweep, which calibrates the highest useful initial temperature per design against the best placement any seed achieves; (3) the retention and quality numbers per seed, with the full-seed versus inter-cluster-only split, which measures sensitivity to the intra-cluster order; the coarse-versus-fine question remains open.

II. Background

A. Simulated annealing placement and RLPlace

VPR places by simulated annealing from an analytic initial placement [5], [6]. The anneal's moves are mostly random; its early phase can undo any structure the start carried. RLPlace improves the anneal rather than the start: a K-armed bandit selects among directed move types (simple moves and swaps, bounded centroid, median, critical-path moves), and checkpointing periodically saves the best placement so oscillation cannot throw it away [4]. Our experiment keeps the anneal as-is (VPR already ships RLPlace-style checkpointing, on by default) and changes the start instead; the checkpointing prediction – that save-best/restore extends the useful temperature range – is one of the hypotheses we test.

B. The seeds

The provenance hierarchy (Urd) is the module structure Odin III records on every netlist object; it needs no recovery. Urd has a single cluster only on sha and stereovision3, two on diffeq1 and diffeq2 (their one module plus a flat group), and one per populated module plus a flat group (3 to 23) on the other designs, so its signal is degenerate on the two one-cluster designs. The recovered clusters

(Verdandi) are Loki III L2 partitions: Louvain, spectral, affinity propagation and k-means at several k, scored in the Loki III paper [7] against the known hierarchy. Agent default: Norns builds the Loki L0 graph.json from the same Odin III BLIF, because Loki's key.json uses Odin API names that do not join to VPR atom names; the clustering code is Loki's. The two seed families differ exactly in the question's terms: provenance is the structure the designer wrote, clusters are the structure the flat netlist still shows.

III. Methodology

A. Seed construction

Each seed is a partition of the placed blocks. Its die-level arrangement comes from the quotient graph: contract each cluster to a node, weight edges by inter-cluster net cuts, lay the quotient out by a coarse force-directed solve, and scale it to the die. Inside a cluster, blocks take the seed's internal order, or a shuffle in the inter-cluster-only arm; the two arms differ only in that intra-cluster order, not in the coarse cluster layout.

B. Seed construction in detail

One die, one netlist. Every seed targets the same die: VPR's k6_frac_N10_frac_chain_mem32K_40nm architecture with six-input LUTs (K = 6). The netlist is the VTR-dialect BLIF that Odin III's front end writes for each design: read_verilog –nolatches –vtr-compat, proc, fsm, opt, flatten, opt -full, wreduce, hardmap, lower, abc –lut 6, stats, and write_blif –vtr –lut-k 6 –name-style provenance. Each configuration is one deterministic VPR run (–seed 1). The unseeded baseline run is the comparison point and also supplies the seed builders: its analytic initial placement (initial.place) gives every soft seed legal slots and its fine order, and its final.flat_place is the reference the retention metric uses.

Coarse layout. Every soft seed shares one die-level arrangement, the quotient-graph layout. Each cluster is contracted to one node and each pair of clusters is weighted by the number of nets cut between them; the weighted quotient graph is laid out by a force-directed solve (networkx.spring_layout, fixed seed), and the embedding is scaled onto the die's logic-slot bounding box. Each cluster then takes the free CLB slots nearest its laid-out point, nearest first. Clusters that exchange many nets therefore start close together, and the die fills in cluster-sized groups instead of a scatter.

Fine order. The soft arm has two variants that differ only inside a cluster. Full keeps the baseline analytic order: a cluster's blocks keep their relative analytic positions when they are assigned the cluster's slots. Inter shuffles the intra-cluster order under a fixed shuffle seed, so only the inter-cluster structure survives; because the coarse cluster layout is identical in the two variants, the pair isolates the fine order. The hard arm has no fine-order choice: it hands VPR one PartitionRegion per cluster, sized from the full seed's slots and expanded until it holds the cluster's logic (and nudged to cover a memory or multiplier site the cluster needs), and VPR's placement orders the blocks inside each region.

Placement macros. Every seed honours the die's placement macros: pad CLBs never move (their I/O sites are fixed by the architecture), mult_36 macros stay anchored on their tile footprint, and an adder carry chain moves as one vertical run in a single column, head at the top (VPR places a chain as a direct vertical macro). Chains are assigned slots before the nearest-first fill, so a chain is never split.

The seed families. The Urd seed is the provenance hierarchy made explicit: Odin III writes each block's HDL module into its netlist name, the seed builder reads that module from the BLIF, and each populated module becomes one cluster; blocks that carry no module name join one flat cluster. Urd is one cluster on sha and stereovision3, two on diffeq1 and diffeq2 (one module plus the flat group), and one per populated module plus the flat group on the other designs, 3 to 23 clusters. The rationale is the designer's locality: modules are the units the designer grouped, and logic inside a module was written to communicate (shared words, state, control), so a module cluster concentrates the connectivity the placer is paid to shorten. That is the hypothesis the Urd arm tests.

The Verdandi seeds are Loki III's L2 clusterings of the flat netlist: Louvain, spectral, affinity propagation, and k-means at k, $2k$ and $4k$, with k the design's module count (two where the design has one module). Norns builds the L0 graph.json the clusterings read from the same Odin III BLIF, because Loki's key.json uses Odin API names that do not join to VPR atom names; the clustering code is Loki's. A recovered cluster approximates the provenance grouping from connectivity alone, so comparing Urd with Verdandi isolates what the HDL structure adds over the graph.

Why a seed should beat a random start. A random placement scatters communicating logic across the die; the quotient layout puts each cluster's logic together and puts clusters that exchange nets next to each other, so the anneal starts from the structure it is meant to find and refines from there. The experiment tests that mechanism rather than assuming it, and it compares against VPR's analytic start, which already respects connectivity, not against a random placement. Two measurements report what the seed did: retention, the fraction of a seed's CLBs found inside its seed cluster's region at exit, has medians 0.68 (full) and 0.70 (inter) at VPR's own start temperature and 0.52–0.74 per family at twice the crossover; and the crossover sweep finds the largest tested temperature scale at which the seed still wins, the point where its contribution stops surviving the anneal.

What the HDL constructs say about a good start. HDL modules, buses, datapath words and finite-state machines group logic that communicates: a bus is one word fanned

out to every consumer, a datapath chain is a dataflow line, and an FSM's next-state logic and registers exchange signals every cycle. Provenance preserves that grouping in the netlist, so a seed built from it starts the placer with the designer's locality already on the die; L2 clusters rediscover most of it from connectivity alone. That is why Urd and Verdandi are the arms, and why the paper asks whether recovered structure is interchangeable with provenance.

C. Injection

Seeds enter VPR two ways, both already in the pinned build (v8.0.0+18416): soft, as an initial placement file (–read_initial_place_file) that the anneal is free to move away from; and hard, as floorplanning constraints (–read_vpr_constraints) with one PartitionRegion per cluster, which the anneal cannot violate at any temperature. The hard arm constrains the regions only, so the anneal chooses the fine placement inside each one (one hard variant, no full/inter split). –write_initial_place_file dumps VPR's own analytic start for the baseline study. VPR's placement macros are respected: I/O pads and the multiplier tiles stay anchored, and carry chains move as one vertical run. The seeds are built from an Odin III provenance BLIF and Loki III's clusterings of the same netlist, arranged by a cut-weighted quotient-graph layout with networkx (spring layout, fixed seed). Building the seed files also needs VPR's unseeded baseline placement for legal slots, so each design's baseline run precedes its seeds.

D. The crossover sweep

The calibration loop follows the data, in order. First, run every seed and keep the best final placement P^* (and which seed made it). Second, cold-start a fresh anneal from P^* and sweep the initial temperature upward; the crossover is the largest tested scale at which the cold start still routes shorter than the unseeded run at the same scale, a lower bound on the true fall point when the sweep stops at its cap. The sweep is run once per design, so S is design-level and identical across that design's seeds. The temperature is reported as a multiple S of VPR's automatic start-temperature estimate. Third, run every seed at twice the design's crossover against an unseeded run at the same temperature: whether the seed still wins there tests how much of its advantage survives a hot restart. The checkpointing arm repeats the sweeps with VPR's checkpointing disabled; per RLPlace, the crossover should move hotter when it is on.

E. Metrics and data

Per run: seed retention (fraction of cluster CLBs still inside their seed cluster's region at exit; mean displacement from the seed coordinate), routed wirelength, critical-path delay, minimum channel width, placement runtime, and the per-temperature cost curve. Designs are the 15 VTR benchmarks that fit the development machine, smallest

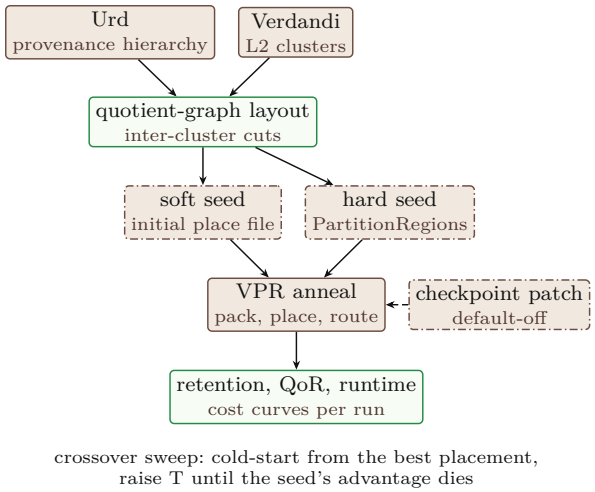


Fig. 1. The Norns III experiment. Seeds from provenance and recovered clusters, arranged by a quotient-graph layout, enter VPR softly (initial placement) or hard (constraints). The crossover sweep cold-starts from the best placement and raises the initial temperature until the seed's advantage dies or the sweep reaches its cap; the checkpointing arm (VPR's stock flag, on vs off) tests whether save-best/restore extends the range.

first, one run at a time under the heavy lock. Every number in this paper traces to a kept record; large data keeps its scripts.

F. Benchmarks

Table I reports each design's construction and its post-technology-mapping size. The module count is the front end's, while the CLB count and the grid are VPR's own numbers for the packed netlist, so the table describes the designs as the experiment actually places them.

The 15 designs span two orders of magnitude after technology mapping: from 13 CLBs (ch_intrinsics) to 1956 (stereovision2), 4843 CLBs in total. The mean is 322.9 CLBs and the median 125; the mean sits well above the median because three stereovision designs carry most of the logic. VPR auto-sizes a square grid to the packed netlist, with a mean side of 27.2 tiles (median 20) and a mean of 1076.4 grid tiles (median 400). Grid size follows the largest resource demand, not the CLB count: I/O-heavy and memory-heavy designs such as mkDelayWorker32B get a large die for few CLBs.

IV. Results

The 15 completed designs give the ratios of Table II. Soft seeds stay within a few percent of the unseeded baseline in routed wirelength (1.009 for the full seed, 1.015 for the inter-cluster-only arm, both pooled over all seven seed families) while cutting the minimum channel width to 0.973 and costing 1.087 in placement runtime (both also pooled). The hard arm is worse on average (geomean 1.088 over the 104 runs that completed; 14 of those still beat the baseline, and stereovision3's hard geomean is 0.906), but not uniformly: one blob_merge k-means$_{2k}$ run

TABLE I
Benchmark construction and post-technology-mapping size. All fifteen designs are from the VTR 8.0 benchmark set (vtr_flow/benchmarks/verilog). Modules is the front end's module count, CLBs the clb blocks VPR packs, Memory and Mult the memory and mult_36 blocks, and Grid the auto-sized device grid with its tile count in parentheses. Sizes are read from each design's baseline run log and kept in data/benchmark-sizes.json.

Design	Suite	Modules	CLBs	Memory	Mult	Grid (tiles)
ch_intrinsics	VTR 8.0	2	13	1	0	10×10 (100)
stereovision3	VTR 8.0	1	15	0	0	7×7 (49)
diffeq2	VTR 8.0	1	22	0	5	16×16 (256)
diffeq1	VTR 8.0	1	37	0	5	16×16 (256)
mkPktMerge	VTR 8.0	7	21	15	0	28×28 (784)
raygentop	VTR 8.0	15	143	1	6	19×19 (361)
sha	VTR 8.0	1	142	0	0	17×17 (289)
boundtop	VTR 8.0	13	54	1	0	15×15 (225)
mkSMAdapter4B	VTR 8.0	8	125	5	0	20×20 (400)
or1200	VTR 8.0	16	252	2	1	27×27 (729)
stereovision0	VTR 8.0	25	707	0	0	33×33 (1089)
stereovision1	VTR 8.0	16	788	0	43	40×40 (1600)
mkDelayWorker32B	VTR 8.0	16	74	43	0	50×50 (2500)
stereovision2	VTR 8.0	24	1956	0	192	82×82 (6724)
blob_merge	VTR 8.0	2	494	0	0	28×28 (784)

TABLE II
Geomean quality ratios vs the unseeded baseline, per seed family, for the soft initial-placement arm; the inter-cluster-only row shuffles the order inside each cluster and pools all seven seed families, and the k-means row pools k, $2k$ and $4k$. Below 1.0 is better. The hard-constraint arm is discussed in the text.

Seed	Wirelength	Delay	Chan. width	Runtime
Urd (hierarchy)	1.029	1.004	0.958	1.068
Louvain	1.008	0.982	0.980	1.091
Spectral	1.006	0.993	0.968	1.098
Affinity propagation	1.002	1.003	0.982	1.086
k-means (k, $2k$, $4k$)	1.006	0.985	0.975	1.088
Inter-cluster only	1.015	0.981	0.987	1.102

TABLE III
The crossover sweep, per seed family: the largest tested scale S of VPR's automatic start-temperature estimate at which the cold start from P^* (the best placement over all seeds) still routes shorter than an unseeded run at the same S, and the seed's median retention at the hot step ($2S$). S is design-level (one sweep per design), so it is identical across families; where the sweep stopped at its cap, S is a lower bound.

Seed	Crossover S	Retention
Urd (hierarchy)	16×	0.744
Louvain	16×	0.520
Spectral	16×	0.635
Affinity propagation	16×	0.564
k-means (k, $2k$, $4k$)	16×	0.667

is missing from the arm because VPR's initial placement could not fit 9 CLBs inside its regions (the constraints are too tight, not a packing failure). Fixed regions may block the anneal's fine moves across cluster borders – a hypothesis the aggregate numbers are consistent with. No single seed family dominates: Urd is the worst full seed in soft wirelength (1.029) and the best hard seed (1.062 against 1.075-1.120 for the recovered families). Retention at scale 1 is 0.68 of a seed's cluster CLBs (full) and 0.70 (inter).

V. Discussion

Three observations. First, the seed families span a narrow band but are not interchangeable: provenance (Urd) is the worst full seed in the soft arm (1.029) and the best in the hard arm, so it is not consistently better or worse than recovered clusters. The full and inter-cluster-only arms differ by 0.6% in pooled wirelength (1.009 vs 1.015); no arm randomizes the coarse cluster layout, so whether the fine order inside a cluster matters as much as the coarse die position is not measured here and stays open. Second, VPR's checkpointing does not separate the crossover in the measured range: the two sweeps find the same scale on 14 of the 15 designs where both found one, and the control extends it on stereovision0 (control 32 vs 16). RLPlace's prediction that checkpointing extends the useful range is therefore not observed on these designs; if it holds, its effect is smaller than the between-design variation. Third, the safe recipe is to cold-start from a cluster-derived placement at a temperature scale at or below the crossover, which is a multiple of VPR's own estimate; above it the seed's retention is still high per family (medians 0.52-0.74 at twice the crossover, individual runs 0.21-1.00) but it wins only 57 of 105 head-to-heads against an unseeded run at the same temperature – the coarse structure survives the anneal, the advantage does not.

VI. Conclusion

Seeding VPR's anneal is worth it only as a calibrated cold start: at VPR's own start temperature the cluster layouts and the provenance hierarchy place at parity with the analytic start, but from a good final placement they keep winning when the anneal is restarted hotter, up to the largest tested scale that still wins – a lower bound on the fall point where the sweep hit its cap. The recipe

is that crossover table, not a seed-specific tuning: start at or below it, and expect the seed to inform the coarse placement while the anneal supplies the fine one.

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